

Colorado Avalanche Fatalities

How avalanche fatalities are evolving over the past 75 Years. An examination of Colorado Avalanche Information Center accident data.

Q1 - National Trend: how fatalities have changed over 75 seasons nationally.

Q2 - Human Factors: which activities are most associated with fatal accidents, and how that mix has shifted.

Q3 - Seasonality & Geography: when and where accidents cluster, Colorado vs. the rest of the nation.

Q4 - Multi-Fatality Events: how often a single avalanche kills more than one person.

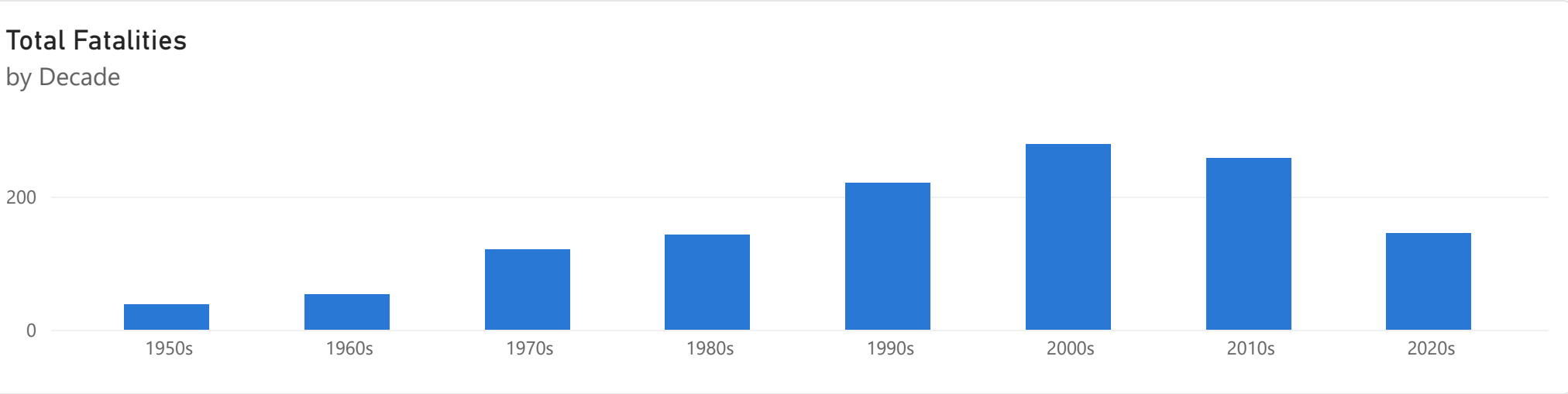
Q5 - Readiness vs. Fatalities: does rising engagement with CAIC's safety information track with fewer deaths?

Data Quality & Methodology: an honest look at this dataset's limitations.

Q1: National Trend

Avg Fatalities Per Season

16.73



Caveat: 1950s (9 seasons) and 2020s (6 seasons, through 2025) are partial decades, not directly comparable to full ones. Record-keeping also improved over time (see Data Quality page), so part of the long-run rise may reflect better documentation, not just more real fatalities.

Deadliest Season

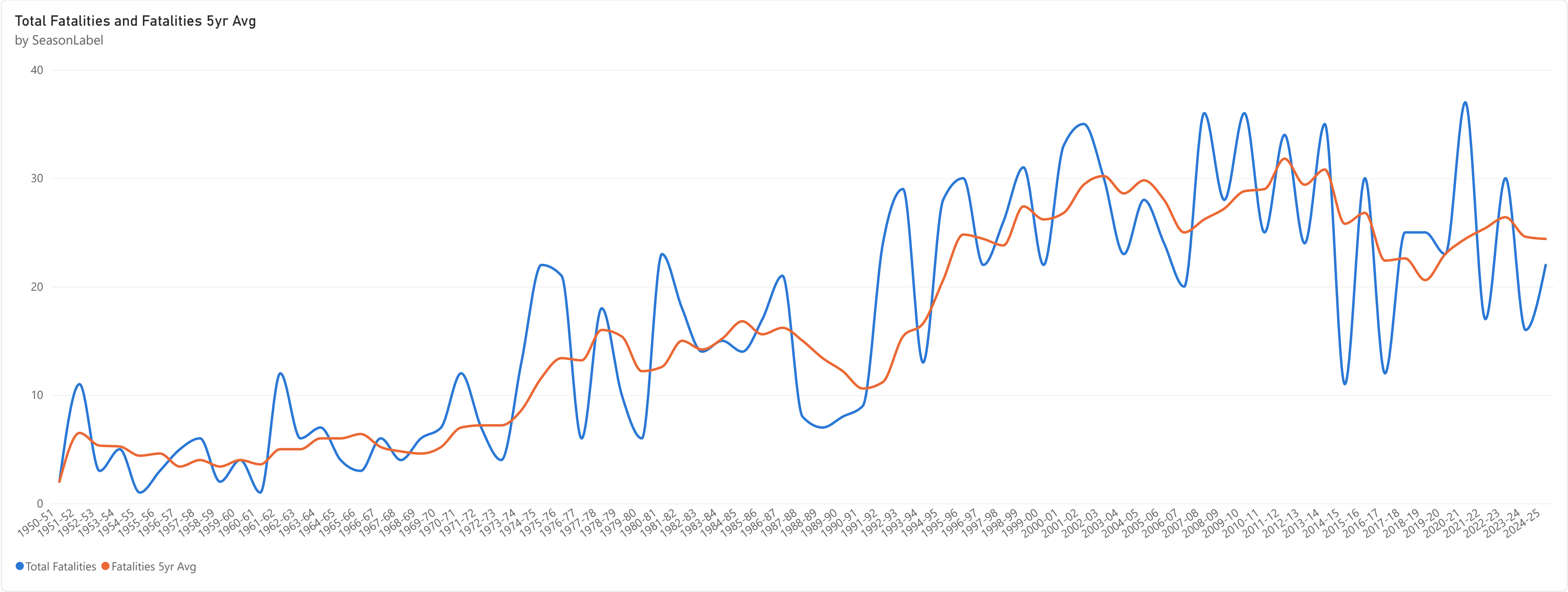
▼

2020-21

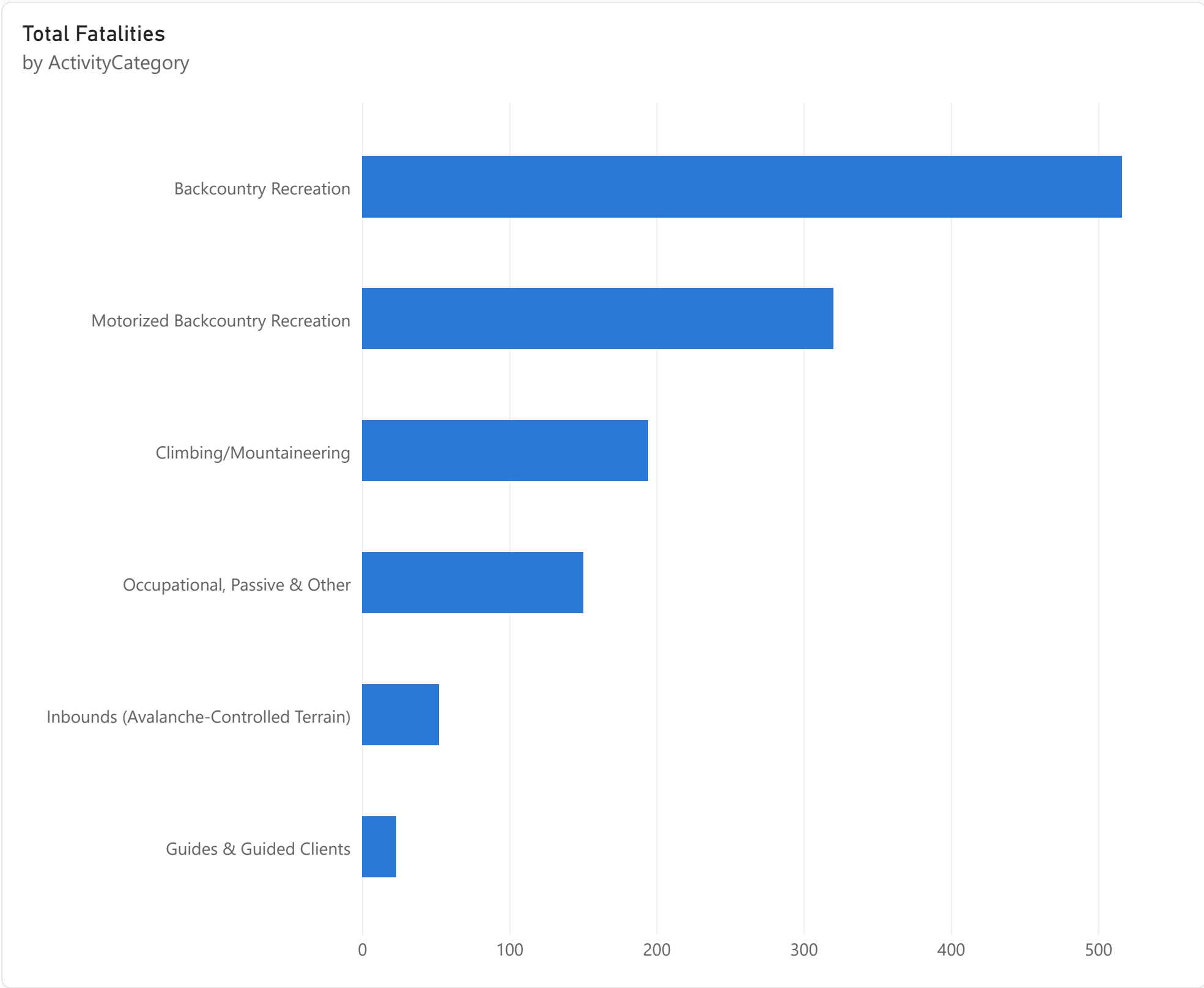
Deadliest Season Fatalities

37

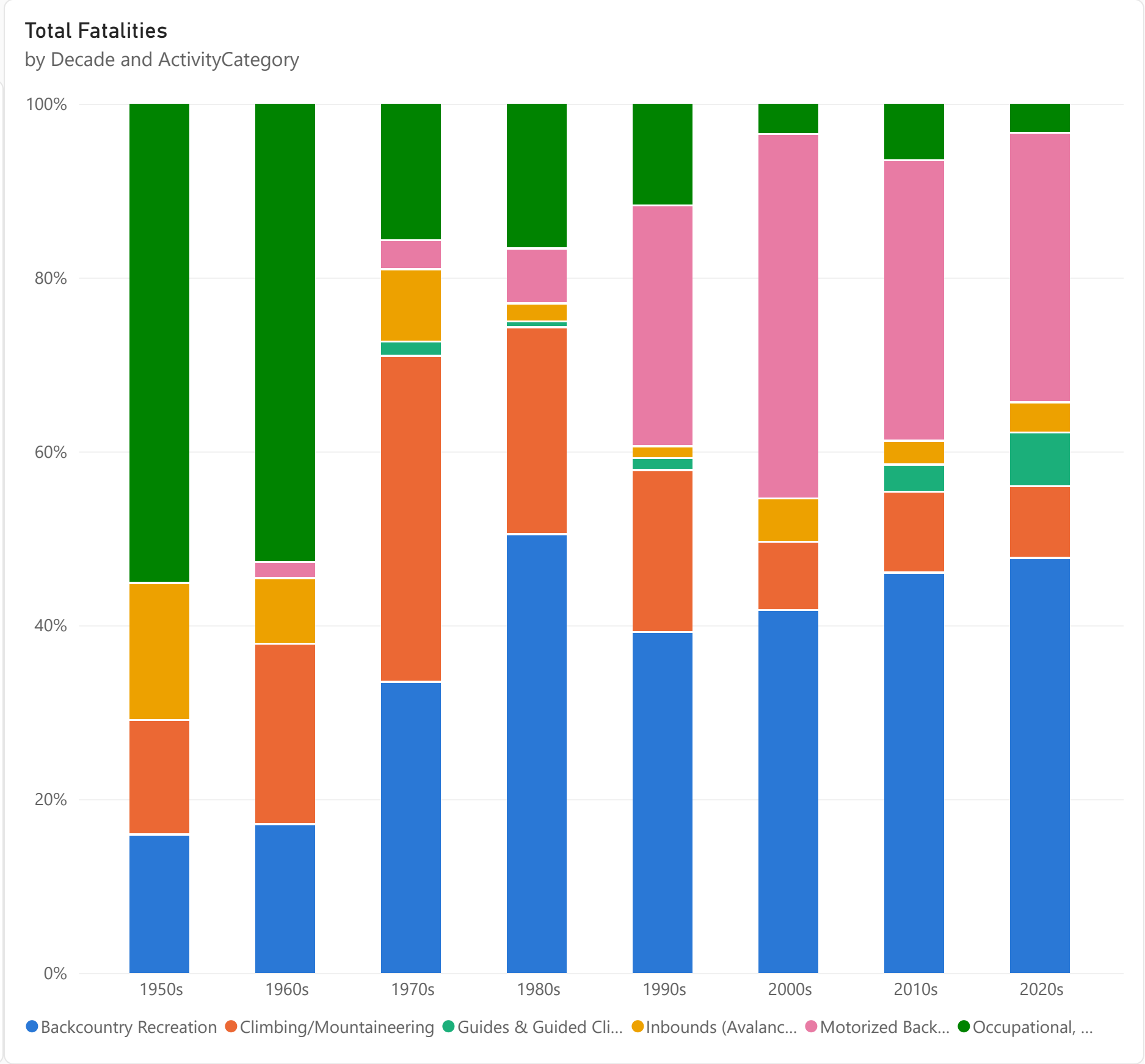
Finding: Fatalities rose sharply from the 1950s through the 1990s (~4 → ~22 per season, a 5x increase), then flattened — the last three decades all sit in a 24-28/season band. The outlier is 2020-21, the deadliest season on record (37 deaths), likely driven by a pandemic-era surge of inexperienced backcountry users.



Q2: Human Factors & Modern-era Activity Breakdown

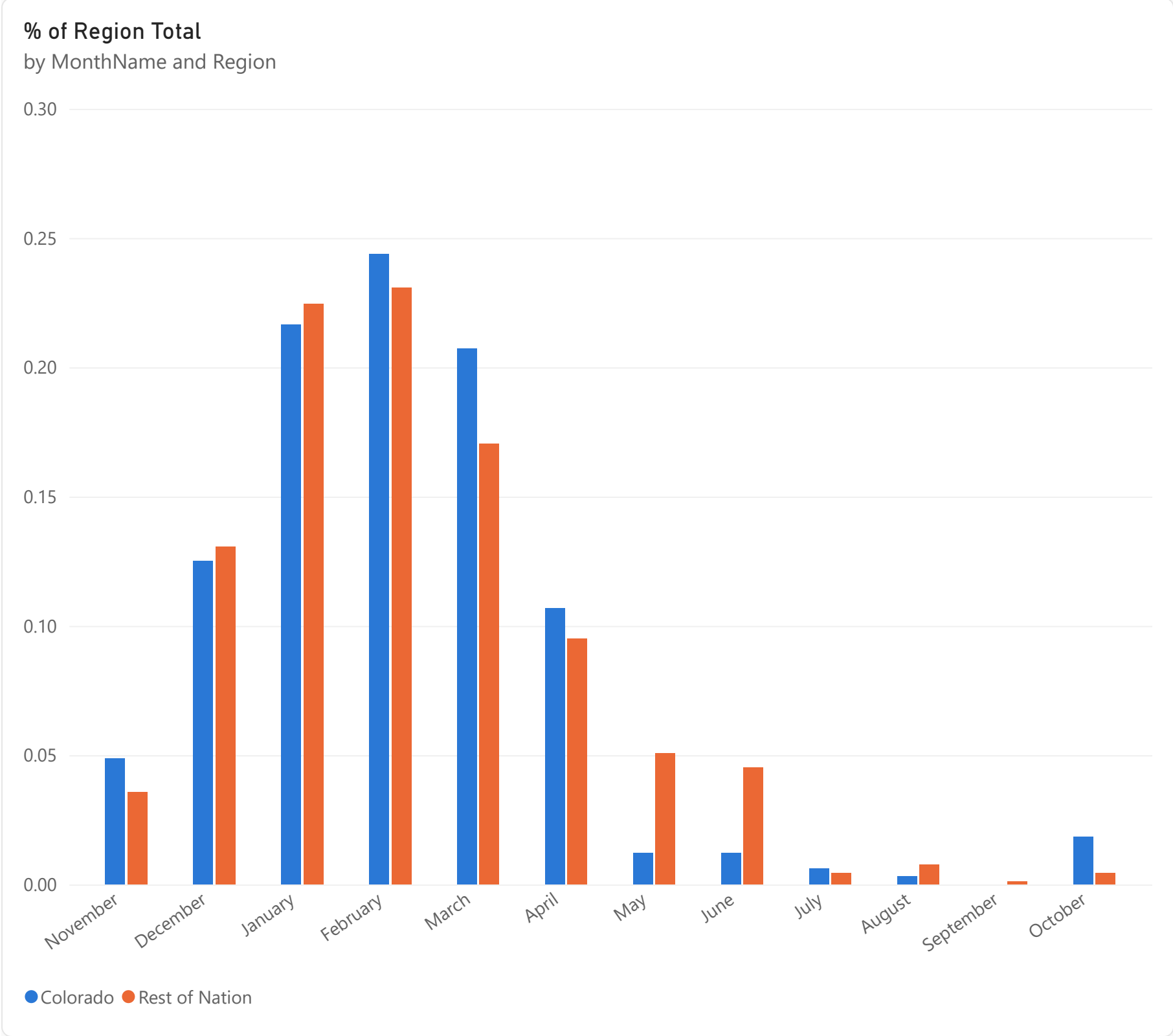


Caveat: "Occupational, Passive & Other" is a mixed category — it combines people doing their job in avalanche terrain (patrol, highway crews), people the avalanche reached passively (residents, motorists), and small leftover recreational activities. Its position on this chart reflects total count only, not a shared cause or story. Activity categories reflect each investigator's judgment at the time, not a standardized taxonomy applied consistently across 75 years (see data dictionary). The 1950s and 2020s are also partial decades.

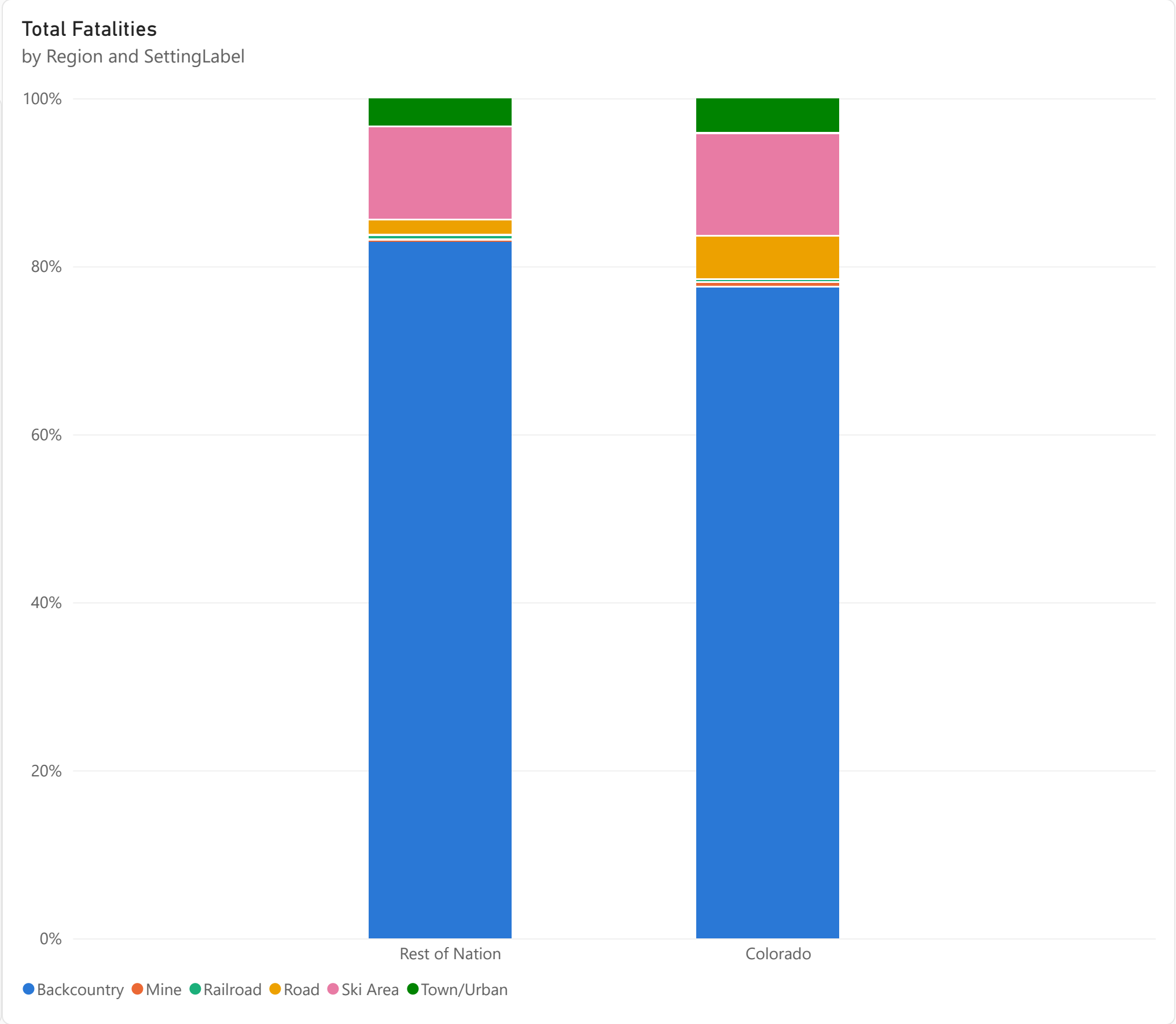


Finding: Two structural shifts stand out. Through the 1950s–60s, most deaths were occupational or passive — road crews, miners, residents with no choice about being in avalanche terrain — over half of all fatalities in those decades, falling below 10% since the 2000s as voluntary recreation became the norm. The other shift is access mode: motorized backcountry recreation, essentially absent before the 1990s, grew to over 40% of fatalities by the 2000s, tracking the real snowmobile-access boom — still one of the two largest categories today.

Q3: Seasonality and Geography - Colorado in National Context



Caveat: Setting and other categorical fields reflect each investigator's judgment, not a standardized taxonomy — and Colorado is the only state CAIC staff investigated directly; other states' incidents were only compiled from other sources. That said, don't read this as "any CO difference is probably noise" — Colorado's continental snowpack is genuinely known for persistent, deep instability unlike faster-stabilizing maritime snowpacks, so real geographic differences are expected here, not just data artifacts.

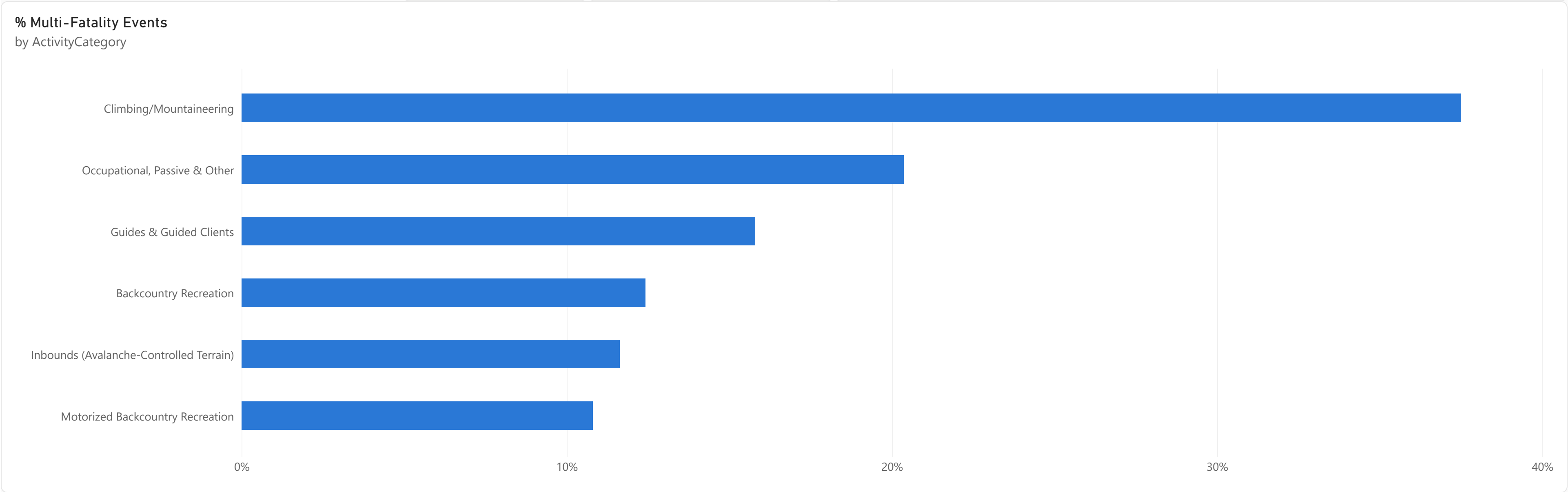


Finding: Timing: both regions peak Feb–March, but Colorado's season runs hotter into March (20.7% of its total vs. 17.0% nationally) while the rest of the nation has a longer late-spring tail (May–June: 9.6% vs. 2.4%), likely reflecting high-elevation glaciated terrain in Alaska and Washington. Setting: Road incidents are roughly 3x Colorado's share of the rest of the nation's (5.2% vs. 1.8%) — plausibly its mountain highway corridors — though this rests on a modest count (17 either way).

Q4: Multi-Fatality Group Events

% Multi-Fatality Events	% Fatalities from Multi-Fatality Events
16%	32%

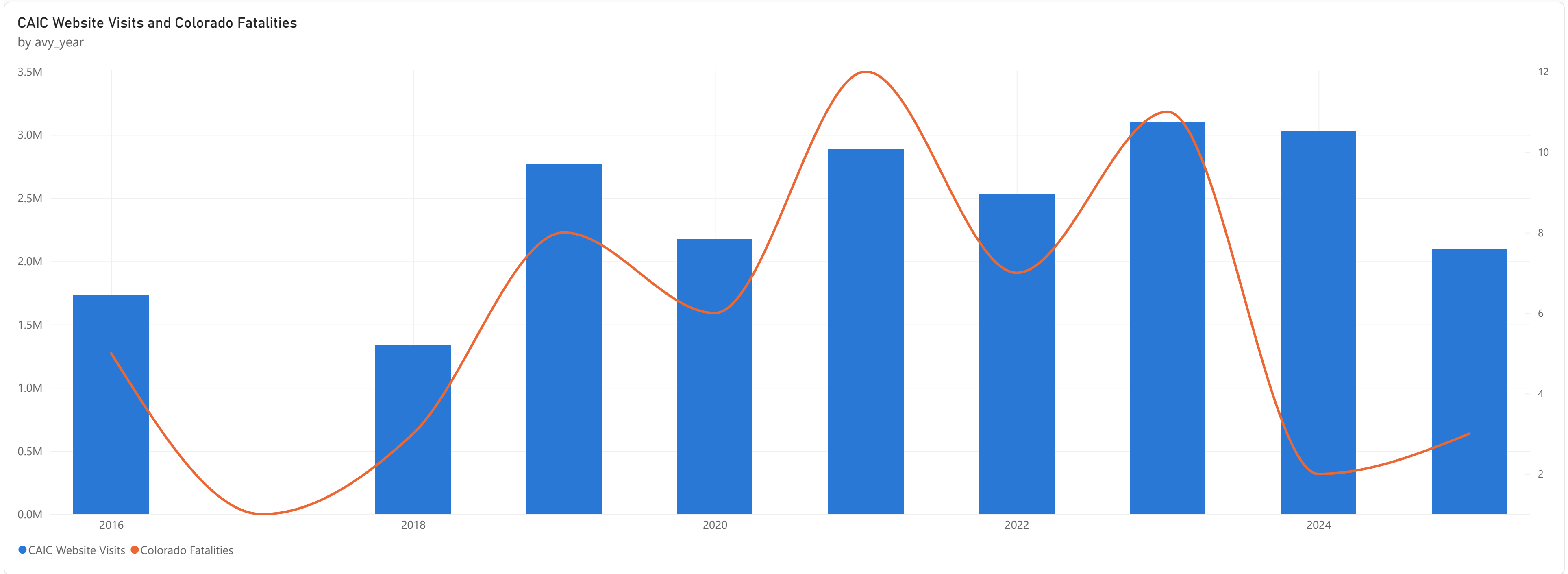
FatalityGroupSize	Avg People Caught (Modern)	Modern Parsed Events	Avg Killed (Modern, Parsed)	Survival Rate (Modern)
Single-Fatality (1)	1	234	1	0.32
Multi-Fatality (2+)	3	27	2	0.24
Total	2	261	1	0.30



Caveat: The severity table covers 261 modern-era (2013-2025) events with a valid People Caught parse — only 27 are multi-fatality, a small sample to read directionally. Activity categories reflect individual investigator judgment, not a standardized taxonomy (see Q2/Q3).

Finding: Multi-fatality events are 16% of all fatal accidents but 32% of all deaths. People caught in one also survive less often — 24% in the modern era, vs. 32% for single-fatality events. Climbing/Mountaineering has the highest multi-fatality rate of any category (37.5%, vs. 11-20% elsewhere), likely reflecting roped travel, where one avalanche can catch an entire party.

Q5: CAIC Engagement vs. Colorado Fatalities

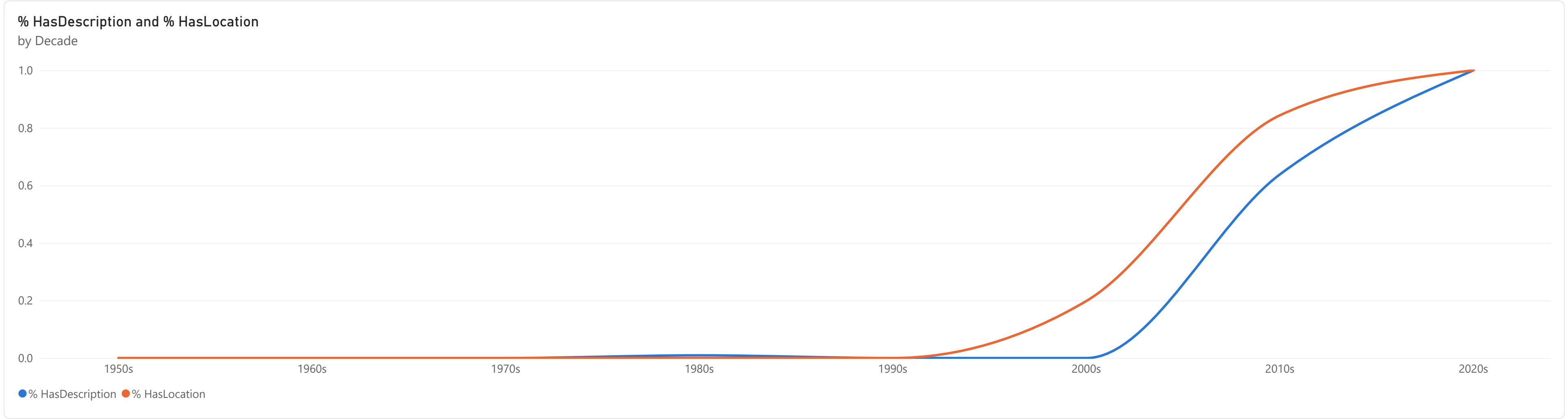


Caveat: This measures engagement with CAIC's safety information, not a literal participant headcount — it can't distinguish one person checking daily from many people checking once, and some traffic is likely storm- or news-driven rather than trip planning. The metric itself changes definition mid-series (website traffic switched from 'visits' to 'pageviews' between FY17 and FY18, not directly comparable), and the 2017 gap in the chart is a known duplicate value in CAIC's own report rather than missing data. FY21-22's fatality count is backfilled rather than report-stated, and 10 seasons is enough to see a direction, not enough to prove one. A peer-reviewed 2022 study using CAIC's own fatality data tried to track avalanche-education trends over a similar window and had to drop the analysis entirely — 78% of the underlying reports didn't record it.

Finding: Website engagement and Colorado fatalities move largely independently of each other over this decade. Engagement has a real upward relationship with time (moderate positive correlation, $r \approx 0.57$, excluding the flagged FY16-17 value): roughly 40% higher in the most recent three years than the first three. Fatalities show almost no relationship to time at all ($r \approx 0.16$), bouncing up and down mainly with how dangerous each season's snowpack was rather than tracking engagement. That pattern fits with the idea that as more people prepare for backcountry trips, risk per person is going down, even though we can't say for sure why — better gear, more education, and more cautious decisions could all be playing a part.

Data Quality and Methodology Caveats

This is NOT a findings page - this page is designed to show the limitations of this dataset



Location data became reliably recorded starting in the 2000s; narrative descriptions only starting in the 2010s — both reached 100% by the 2020s.

Investigator judgment: Setting, PrimaryActivity, and other categorical fields reflect individual investigators' judgment at the time, not a standardized taxonomy applied consistently across 75 years (confirmed by CAIC's Spencer Logan). An apparent shift in the data may partly reflect changing investigator conventions, not just changing conditions on the mountain.

Colorado vs. rest of nation: Colorado's own rows were investigated directly by CAIC staff; incidents in other states were only compiled from other sources. Colorado's data is the most internally consistent subset in this dataset — a reason to trust Colorado-specific comparisons (like Q3) more than nationwide ones.

Engagement metric changes: The CAIC engagement proxy used on Q5 changes its own definition mid-series (visits → pageviews → users), has one suspected duplicate data point (FY16-17), and one backfilled fatality count (FY21-22) rather than report-stated. It's directionally useful, not a single clean line.

2025 is provisional: CAIC's own raw data flags 2024-25 as still 'Current Year' as of this export — not a closed, finalized season. Any total or chart including AvyYear 2025 should be read as provisional, not final.